Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(d)		$n = \frac{pV}{RT} = \frac{115\times10^{3}\times2.2\times10^{-3}}{8.31\times294}$ = 0.104 [mol] (1) Final temperature, $T = \frac{pV}{nR} = \frac{115\times10^{3}\times2.6\times10^{-3}}{0.104\times8.31}$ = 346.0 K (1) [value depends upon rounding of n. No rounding $\rightarrow$ 347.5 K] $\Delta U = \frac{3}{2}nR\Delta T = \frac{3}{2}\times0.104\times8.31(346-294)$ = 67.4 J (1) Allow ecf on 1 mol for the last two marks. **Alternative:** $\Delta U = \frac{3}{2}(p_1V_1 - p_2V_2)$ or $\Delta U = \frac{3}{2}p\Delta V$ (for constant pressure) (1) $= \frac{3}{2}115\times10^{3}(2.60-2.20)\times10^{-3}$ (1) = 69 J (1)		3		3	3	
			Question 2 total	**9**	**3**	**0**	**12**	**5**	**0**